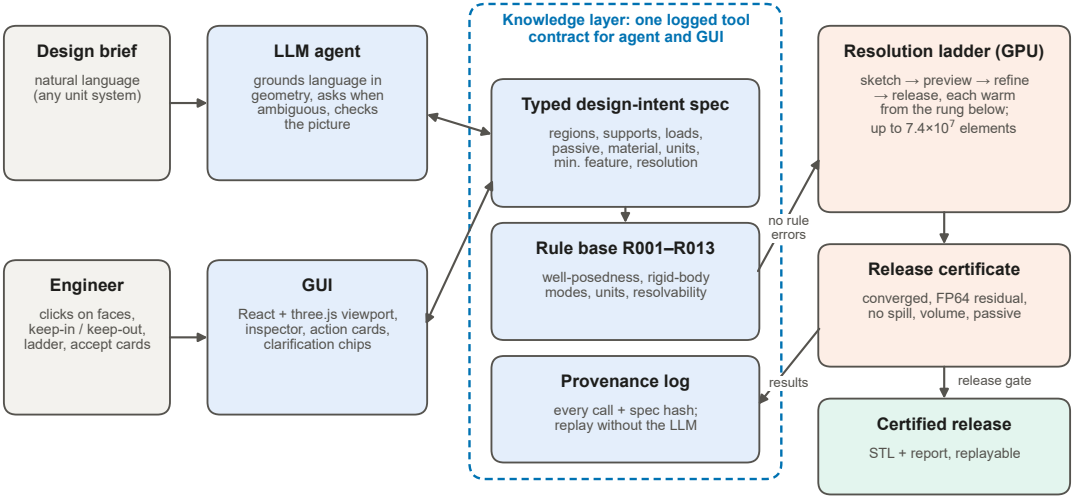
Design brief
natural language
(any unit system)
LLM agent
grounds language in
geometry, asks when
ambiguous, checks
the picture
Engineer
clicks on faces,
keep-in / keep-out,
ladder, accept cards
GUI
React + three.js viewport,
inspector, action cards,
clarification chips
Knowledge layer: one logged tool
contract for agent and GUI
Typed design-intent spec
regions, supports, loads,
passive, material, units,
min. feature, resolution
Rule base R001–R013
well-posedness, rigid-body
modes, units, resolvability
Provenance log
every call + spec hash;
replay without the LLM
no rule
errors
results
Resolution ladder (GPU)
sketch → preview → refine
→ release, each warm
from the rung below;
up to 7.4×10^7 elements
Release certificate
converged, FP64 residual,
no spill, volume, passive
release gate
Certified release
STL + report, replayable